

Multiple Choice

1. **Answer:** C

Options: A) retraction of the stoma.; B) high output stomas.; C) herniation around the stoma.; D) infection around the stoma.; E) dehydration caused by the stoma.

Note: A surgical support is designed to provide stability and protection to areas vulnerable to herniation, particularly around stomas, where the abdominal wall may be weakened.

2. **Answer:** C

Options: A) A black woman; B) A Hispanic man; C) A white woman; D) An Asian man; E) A white man

Note: Research consistently shows that demographic factors such as race and gender influence longevity, with white women generally having higher life expectancy rates due to a combination of biological, social, and healthcare access factors.

3. **Answer:** A

Options: A) The patient has no difficulty in breathing.; B) It is 24 hours after the tracheostomy tube was first inserted.; C) The patient is able to speak clearly.; D) The patient's vital signs are stable.; E) That the patient is not on a ventilator.

Note: Before using a speaking valve, it is essential to ensure that the patient has no difficulty in breathing because the valve can impede airflow, potentially leading to respiratory distress if the patient's airway is compromised.

4. **Answer:** A

Options: A) 20-30 minutes.; B) 1-2 seconds.; C) 10-15 minutes.; D) 30-40 seconds.; E) 90-100 seconds.

Note: The correct answer is 20-30 minutes because the energy stored in ATP and phosphocreatine is sufficient to support high-intensity activities for this duration before the body transitions to longer-term energy systems, such as anaerobic glycolysis and aerobic metabolism.

5. **Answer:** A

Options: A) Decreased muscle mass; B) Impaired digestion; C) Worsening of vision; D) Increase in blood sugar levels; E) Reduction of bone density

Note: Decreased muscle mass is not a beneficial effect of exercise in people with diabetes because regular physical activity is known to enhance muscle strength and mass, which can improve insulin sensitivity and overall metabolic health.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The descending loop of Henle is permeable to water but not to ions, allowing for passive reabsorption of water and concentration of urine, while the presence of H^+ ions in the surrounding interstitial fluid can influence water movement, thus supporting the statement's validity in the context of overall ion exchange processes in the nephron.

7. **Answer:** True

Options: A) True; B) False

Note: Cognitive symptoms in Parkinson's disease typically manifest later in the disease progression and are often less severe compared to the prominent and early cognitive decline observed in other forms of dementia, such as Alzheimer's disease.

8. **Answer:** False

Options: A) True; B) False

Note: Research indicates that factors such as socioeconomic status, access to health-care, and lifestyle choices, rather than the number of children a woman has, have a more significant impact on women's longevity.

9. **Answer:** False

Options: A) True; B) False

Note: Vitamin D is primarily involved in calcium metabolism and bone health, rather than serving as a cofactor in hydroxylation reactions that are essential for collagen synthesis, which primarily requires vitamin C as a cofactor.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because while pulmonary hypoplasia can occur due to oligohydramnios, it is not exclusively classified as a syndrome related to renal agenesis, as oligohydramnios can also result from other conditions unrelated to the kidneys.

Long Form Response

11. **Answer:** Ectomesenchyme, derived from neural crest cells, plays a crucial role in the development of various structures in the body, particularly in the formation of melanocytes. These pigment-producing cells are essential for the synthesis of melanin, which not only gives color to the skin, hair, and eyes but also provides protection against ultraviolet radiation. The significance of melanocytes extends into health sciences, as variations in their function can lead to conditions such as vitiligo, melanoma, and other skin disorders. Understanding the developmental pathways and regulatory mechanisms of ectomesenchyme in melanocyte formation is vital for advancing therapeutic strategies in dermatology and oncology. Thus, ectomesenchyme is integral not only to structural development but also to maintaining skin homeostasis and addressing health challenges associated with pigmentary disorders.

Note: correct because it accurately links ectomesenchyme, which originates from neural crest cells, to the development of melanocytes, highlighting their critical role in melanin production and the implications for health conditions related to pigmentation.

12. **Answer:** The increasing age for receiving full social security benefits has significant implications for public health and individual well-being in contemporary society. As the eligibility age rises, many individuals find themselves working longer or postponing retirement, which can lead to increased stress and mental health challenges, particularly among older adults. Furthermore, this delay in accessing benefits may exacerbate health disparities, as those in lower socioeconomic positions might struggle to maintain their health and employment longer than their wealthier counterparts. Additionally, the extended working years can hinder the ability of older adults to engage in preventive health measures or recovery strategies, ultimately affecting overall public health outcomes. Therefore, the policy shift not only impacts financial security but also poses challenges to the health and well-being of an aging population.

Note: The answer correctly identifies that the rising age for full social security benefits leads to prolonged work and delayed retirement, which can negatively impact mental health and widen health disparities among different socioeconomic groups.